

Let V be a vector space of all continuous functions defined on the interval $[1, 5]$ and $\langle, \rangle$ is a scalar product defined as

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

1. calculate the orthonormal basis of the subspace $\text{Span}\{1, x, x^2\}$ using Gram-Schmidt's orthogonalisation
2. the orthogonal projection of $\sin x$ into this subspace as a polynome $ax^2 + bx + c$ where $a, b, c \in \mathbb{R}$
3. the norm of the difference $\|\sin x - ax^2 - bx - c\|$

The norm of 1 is given by

$$\sqrt{\int_1^5 1 dx} = \sqrt{[x]_1^5} = \sqrt{4} = 2$$

therefore the first vector of the basis is $1/2 \cdot 1$

The orthogonal projection of x into the subspace generated by $1/2 \cdot 1$ is

$$\langle 1/2, x \rangle \cdot 1/2 = \int_1^5 x/2 dx \cdot 1/2 = [x^2/4]_1^5 \cdot 1/2 = (25/4 - 1/4) \cdot 1/2 = 3$$

The norm of $x - 3$ is given by

$$\sqrt{\int_1^5 (x-3)^2 dx} = \sqrt{\int_1^5 x^2 - 6x + 9 dx} = \sqrt{[x^3/3 - 3x^2 + 9x]_1^5} = \sqrt{16/3}$$

therefore the second vector of the basis is $\sqrt{3}/4 \cdot (x - 3)$

The orthogonal projection of x^2 into the subspace generated by $1/2; \sqrt{3}/4 \cdot (x - 3)$ is

$$\begin{aligned} & \langle 1/2, x^2 \rangle \cdot 1/2 + \langle \sqrt{3}/4 \cdot (x - 3), x^2 \rangle \cdot \sqrt{3}/4 \cdot (x - 3) \\ & \int_1^5 \frac{x^2}{2} dx \cdot 1/2 + \frac{\sqrt{3}}{4} \int_1^5 x^3 - 3x^2 dx \cdot \sqrt{3}/4 \cdot (x - 3) \\ & \left[\frac{x^3}{6} \right]_1^5 \cdot 1/2 + \frac{\sqrt{3}}{4} \left[\frac{x^4}{4} - x^3 \right]_1^5 \cdot \sqrt{3}/4 \cdot (x - 3) = 124/6 \cdot 1/2 + \frac{\sqrt{3}}{4} \cdot 32 \cdot \sqrt{3}/4 \cdot (x - 3) \\ & \frac{31}{3} + 6(x - 3) = 6x - \frac{23}{3} \end{aligned}$$

The norm of $x^2 - 6x + 23/3$ is given by

$$\begin{aligned}\sqrt{\int_1^5 (x^2 - 6x + 23/3)^2 dx} &= \sqrt{\int_1^5 ((x-3)^2 - 4/3)^2 dx} = \sqrt{\int_{-2}^2 (y^2 - 4/3)^2 dy} \\ \sqrt{\int_{-2}^2 y^4 - \frac{8}{3}y^2 + \frac{16}{9} dy} &= \sqrt{\left[\frac{y^5}{5} - \frac{8y^3}{9} + \frac{16y}{9}\right]_{-2}^2} = \sqrt{\frac{256}{45}} = \frac{16}{3\sqrt{5}}\end{aligned}$$

therefore the third vector of the basis is $\frac{3\sqrt{5}}{16}(x^2 - 6x + 23/3)$

Final orthonormal basis $B = \left(\frac{1}{2}; \frac{\sqrt{3}}{4}(x-3); \frac{3\sqrt{5}}{16}\left((x-3)^2 - \frac{4}{3}\right)\right)$

To find the orthogonal projection of $\sin x$ into $\text{Span}\{1, x, x^2\}$, we will use the same process as before

$$\begin{aligned}\frac{\langle 1, \sin x \rangle}{||1||^2} &= c \\ \frac{\langle x-3, \sin x \rangle}{||x-3||^2} &= b \\ \frac{\langle (x-3)^2 - 4/3, \sin x \rangle}{||(x-3)^2 - 4/3||^2} &= a\end{aligned}$$

$$\int_1^5 \sin x dx = [-\cos x]_1^5 = \cos(5) - \cos(1)$$

$$\int_1^5 1 dx = 4$$

$$\int_1^5 (x-3) \sin x dx = [\sin x - x \cos x + 3 \cos x]_1^5 = \sin(5) - 2 \cos(5) - \sin(1) - 2 \cos(1)$$

$$\int_1^5 (x-3)^2 dx = \frac{16}{3}$$

$$\int_1^5 \left((x-3)^2 - \frac{4}{3}\right) \sin x dx = \left[2x \sin x - 6 \sin x + 6x \cos x - (x^2 - 2) \cos x - \frac{23}{3} \cos x\right]_1^5$$

$$= \frac{2}{3} \left(6 \sin(1) + 6 \sin(5) + \cos(1) - \cos(5)\right)$$

$$\int_1^5 \left((x-3)^2 - \frac{4}{3}\right)^2 dx = \frac{256}{45}$$

$$a = \frac{9}{128} \left(6 \sin(1) + 6 \sin(5) + \cos(1) - \cos(5)\right) \approx -0,0315$$

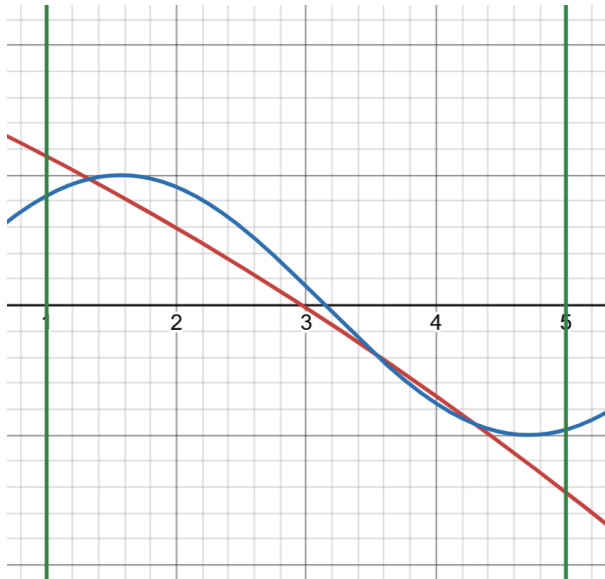
$$b = \frac{3}{16} \left(\sin(5) - 2 \cos(5) - \sin(1) - 2 \cos(1)\right) \approx -0,6466$$

$$c = \frac{1}{4} \left(\cos(5) - \cos(1)\right) \approx -0,0642$$

Where the projection is given by $c(1) + b(x - 3) + a((x - 3)^2 - 4/3)$. Distributed and simplified

$$-0,0315x^2 - 0,4576x + 1,6341$$

Which overlayed on $\sin x$



Lastly the norm of the difference $\sin x + 0,0315x^2 + 0,4576x - 1,6341$ is

$$\sqrt{\int_1^5 (\sin x + 0,0315x^2 + 0,4576x - 1,6341)^2} \approx 0,4124$$

Q. E. D.